Fama–French Five-Factor Model with Time-Varying Loadings

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April 1, 2026

1 Introduction

These notes derive the Fama–French five-factor (FF5) asset pricing model from first principles using the stochastic discount factor (SDF) approach[1]. I begin with the consumer's intertemporal optimization problem and derive the Euler equation, then show how the Euler equation implies a linear pricing relation between expected returns and covariances with the SDF. By approximating the SDF with a linear function of traded factor returns, I obtain the familiar linear factor pricing model and then tailor this framework to the Fama–French five factors. Finally, I extend the model to allow for time-varying factor loadings using both observable state variables (instrumented betas) and a latent state-space (Kalman filter) approach.

2 Setup

Assume that time is discrete and indexed by $t = 0, 1, 2, \ldots$, I consider asset returns between dates t and $t+1$. Let $R_{i,t+1}$ denote the gross return on asset or portfolio i from t to $t+1$. Let $R_{f,t}$ denote the gross risk-free return known at time t. The excess return on asset i is defined as

$$R^e_{i,t+1} \equiv R_{i,t+1} - R_{f,t}.$$

Information available at time t is denoted by $\mathcal{I}_t$. The central object in modern asset pricing is the stochastic discount factor, denoted by m_{t+1}.

In empirical work, and in these notes, I focus on pricing test portfolios rather than individual stocks. Test portfolios are typically formed by sorting stocks on characteristics such as size, book-to-market, profitability, or investment and then averaging returns within each group. This diversification reduces idiosyncratic noise and measures exposure to systematic risk.

3 The Consumer Problem and the Euler Equation

To derive the pricing equation from first principles, consider a representative consumer who lives for two periods, t and $t+1$. At time t, the consumer chooses consumption C_t and a portfolio of assets to maximize expected lifetime utility

$$u(C_t) + \beta\, \mathbb{E}_t \left[u(C_{t+1}) \right],$$

where $u(\cdot)$ is a strictly concave and increasing utility function, and $\beta \in (0, 1)$ is a subjective discount factor.

[1] Notes based on lecture by Prof. Laura Veldkamp (CBS), [Cochrane, 2005], [Fama and French, 2015], [Durbin and Koopman, 2012].

Let the consumer have wealth W_t at time t. For simplicity, normalize asset prices at time t to 1. Let $\theta_{f,t}$ denote holdings of the risk-free asset and let $\boldsymbol{\theta}_t$ denote holdings of risky assets with gross return vector $\boldsymbol{R}_{t+1}$. The budget constraint at time t is

$$C_t + \theta_{f,t} + \boldsymbol{\theta}_t^\top \mathbf{1} = W_t.$$

At time $t+1$, wealth is given by

$$W_{t+1} = \theta_{f,t} R_{f,t+1} + \boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}.$$

Assume that all wealth is consumed at time $t+1$, so $C_{t+1} = W_{t+1}$. The Lagrangian is

$$\mathcal{L} = u(C_t) + \beta \mathbb{E}_t \left[u(C_{t+1}) \right] + \lambda_t \left(W_t - C_t - \theta_{f,t} - \boldsymbol{\theta}_t^\top \mathbf{1} \right).$$

Taking first-order conditions with respect to C_t, $\theta_{f,t}$, and each risky asset i yields

$$u'(C_t) = \lambda_t,$$

$$u'(C_t) = \beta \mathbb{E}_t \left[u'(C_{t+1}) R_{f,t+1} \right],$$

$$u'(C_t) = \beta \mathbb{E}_t \left[u'(C_{t+1}) R_{i,t+1} \right].$$

Dividing both sides of the risky-asset condition by $u'(C_t)$ motivates the definition of the stochastic discount factor.

4 Stochastic Discount Factor

Define the stochastic discount factor as

$$m_{t+1} \equiv \beta \frac{u'(C_{t+1})}{u'(C_t)}.$$

Substituting this definition into the first-order condition gives the fundamental pricing equation

$$\mathbb{E}_t \left[m_{t+1} R_{i,t+1} \right] = 1 \quad \text{for all traded assets } i.$$

This equation is the cornerstone of modern asset pricing. It states that the current price of any asset equals the expected discounted (by the SDF) value of its future payoff.

5 Excess Returns and the Covariance Decomposition

Applying the pricing equation to the risk-free asset implies

$$\mathbb{E}_t \left[m_{t+1} R_{f,t+1} \right] = 1,$$

which in turn implies

$$\mathbb{E}_t [m_{t+1}] = \frac{1}{R_{f,t+1}}.$$

Subtracting the risk-free return from any risky return, I obtain the pricing equation for excess returns:

$$\mathbb{E}_t \left[m_{t+1} R^e_{i,t+1} \right] = 0.$$

Using the identity $\mathbb{E}_t[XY] = \mathrm{Cov}_t(X, Y) + \mathbb{E}_t[X]\mathbb{E}_t[Y]$, I can rewrite this as

$$0 = \mathrm{Cov}_t(m_{t+1}, R^e_{i,t+1}) + \mathbb{E}_t[m_{t+1}]\,\mathbb{E}_t[R^e_{i,t+1}].$$

Solving for the expected excess return yields

$$\mathbb{E}_t[R^e_{i,t+1}] = -\frac{\mathrm{Cov}_t(m_{t+1}, R^e_{i,t+1})}{\mathbb{E}_t[m_{t+1}]}.$$

This expression provides a clear economic interpretation. An asset commands a high expected return if it tends to perform poorly in states where the SDF is high, or equivalently, in bad times when consumption is low and thus marginal utility thereof is high.

6 Linear Factor Representations of the SDF

The SDF is not directly observable. To obtain an empirical model, I approximate the SDF using a small number of traded factor returns. Let $f_{t+1} \in \mathbb{R}^K$ denote a vector of factor excess returns. I assume that the SDF is affine in these factors:

$$m_{t+1} = a_t - b_t^\top f_{t+1},$$

where a_t and b_t may depend on information available at time t. Substituting this expression into the excess return pricing equation gives

$$0 = \mathbb{E}_t\left[(a_t - b_t^\top f_{t+1})R^e_{i,t+1}\right] = a_t\mathbb{E}_t[R^e_{i,t+1}] - b_t^\top\mathbb{E}_t[f_{t+1}R^e_{i,t+1}].$$

Rearranging, I obtain

$$a_t\mathbb{E}_t[R^e_{i,t+1}] = b_t^\top\mathbb{E}_t[f_{t+1}R^e_{i,t+1}].$$

Define the conditional beta vector as

$$\beta_{i,t} \equiv \mathrm{Var}_t(f_{t+1})^{-1}\mathrm{Cov}_t(f_{t+1}, R^e_{i,t+1}),$$

and define the vector of factor risk premia as

$$\lambda_t \equiv \mathrm{Var}_t(f_{t+1})\left(\frac{b_t}{a_t}\right).$$

With these definitions, expected excess returns satisfy the conditional linear factor pricing relation

$$\mathbb{E}_t[R^e_{i,t+1}] = \beta_{i,t}^\top\lambda_t,$$

and a linear SDF implies a linear relationship between expected returns and factor exposures.

7 The Fama–French Five Factors

The Fama–French five-factor model specifies the factor vector as

$$f_{t+1} = \begin{pmatrix} MKT_{t+1} \\ SMB_{t+1} \\ HML_{t+1} \\ RMW_{t+1} \\ CMA_{t+1} \end{pmatrix},$$

where MKT is the market excess return, SMB is the size factor, HML is the value factor, RMW is the profitability factor, and CMA is the investment factor. Each factor is constructed as the return on a diversified long–short portfolio formed by sorting stocks on observable characteristics. For example, the value factor HML is constructed by going long high book-to-market stocks and short low book-to-market stocks, averaged across small and big firms. The profitability and investment factors are constructed similarly. Under the FF5 model, the linear SDF takes the form

$$m_{t+1} = a_t - b_{M,t}MKT_{t+1} - b_{S,t}SMB_{t+1} - b_{H,t}HML_{t+1} - b_{R,t}RMW_{t+1} - b_{C,t}CMA_{t+1}.$$

The corresponding empirical time-series model for a test portfolio i is

$$R^e_{i,t+1} = \alpha_i + \beta_{i,M}MKT_{t+1} + \beta_{i,S}SMB_{t+1} + \beta_{i,H}HML_{t+1} + \beta_{i,R}RMW_{t+1} + \beta_{i,C}CMA_{t+1} + \varepsilon_{i,t+1}.$$

If the FF5 model is correctly specified, then $\alpha_i = 0$ in population for all priced test portfolios.

8 Time-Varying Factor Loadings

The FF5 model is often first introduced with constant factor loadings:

$$R^e_{i,t+1} = \alpha_i + \beta_i^\top f_{t+1} + \varepsilon_{i,t+1}, \qquad f_{t+1} = \begin{pmatrix} MKT_{t+1} \\ SMB_{t+1} \\ HML_{t+1} \\ RMW_{t+1} \\ CMA_{t+1} \end{pmatrix}.$$

In many applications, however, a portfolio's factor exposure changes over time. For instance, a firm's leverage, product mix, or financing constraints can vary across business cycles, and those changes can translate into changing sensitivities to systematic factors. In this section, I develop two standard ways to model time-varying factor loadings ("time-varying betas"): (i) instrumented betas that vary with observable state variables, and (ii) latent betas that evolve as an unobserved state in a state-space model estimated via the Kalman filter. I also work through a numerical example for the instrumented model and present a detailed Kalman filter example.

8.1 Instrumented Time-Varying Betas

8.1.1 Model

Let $Z_t \in \mathbb{R}^p$ be a vector of instruments (state variables) observed at time t. Examples include the term spread, default spread, dividend yield, VIX, inflation, or a recession indicator. The key modeling assumption is that the factor loading vector $\beta_{i,t}$ is an affine (linear plus intercept) function of Z_t:

$$\beta_{i,t} = \beta_{i,0} + B_i Z_t, \tag{1}$$

where $\beta_{i,0} \in \mathbb{R}^K$ is the baseline loading vector and $B_i \in \mathbb{R}^{K\times p}$ governs how exposures vary with the state (in FF5, $K = 5$). Start from the conditional factor model for the test portfolio's excess return:

$$R^e_{i,t+1} = \alpha_i + \beta_{i,t}^\top f_{t+1} + \varepsilon_{i,t+1}. \tag{2}$$

Substitute (1) into (2):

$$R^e_{i,t+1} = \alpha_i + (\beta_{i,0} + B_i Z_t)^\top f_{t+1} + \varepsilon_{i,t+1} \tag{3}$$

$$= \alpha_i + \beta_{i,0}^\top f_{t+1} + (B_i Z_t)^\top f_{t+1} + \varepsilon_{i,t+1}. \tag{4}$$

Now use the identity $(B_i Z_t)^\top f_{t+1} = Z_t^\top (B_i^\top f_{t+1})$. In column form,

$$B_i = \big[\gamma_{i,1}, \gamma_{i,2}, \ldots, \gamma_{i,p}\big], \qquad \gamma_{i,j} \in \mathbb{R}^K.$$

Then $B_i Z_t = \sum_{j=1}^p \gamma_{i,j} Z_{j,t}$, so

$$(B_i Z_t)^\top f_{t+1} = \left(\sum_{j=1}^p \gamma_{i,j} Z_{j,t}\right)^\top f_{t+1} \tag{5}$$

$$= \sum_{j=1}^p Z_{j,t}\, \gamma_{i,j}^\top f_{t+1}. \tag{6}$$

Plug (6) into (4):

$$R_{i,t+1}^e = \alpha_i + \beta_{i,0}^\top f_{t+1} + \sum_{j=1}^p Z_{j,t}\, \gamma_{i,j}^\top f_{t+1} + \varepsilon_{i,t+1}. \tag{7}$$

Equation (7) shows that instrumented time variation is implemented using a linear regression on (i) the original factors f_{t+1} and (ii) interaction regressors $Z_{j,t} f_{t+1}$. With estimates for $\beta_{i,0}$ and $\{\gamma_{i,j}\}_{j=1}^p$, compute the fitted time-t exposure:

$$\widehat{\beta}_{i,t} = \widehat{\beta}_{i,0} + \sum_{j=1}^p Z_{j,t}\, \widehat{\gamma}_{i,j}. \tag{8}$$

The time variation in $\beta_{i,t}$ is completely determined by the observed state Z_t and the estimated interaction coefficients.

8.1.2 Numerical Example

To keep the example simple, I use two factors rather than all five. The logic is identical to FF5. Practically, extend from $K = 2$ to $K = 5$ by adding extra interaction columns. Let the factor vector be

$$f_t = \begin{pmatrix} MKT_t \\ HML_t \end{pmatrix},$$

and let the instrument be a scalar z_{t-1} observed at time $t-1$. Specify time-varying betas as

$$\beta_{M,t-1} = \beta_M + \delta_M z_{t-1}, \qquad \beta_{H,t-1} = \beta_H + \delta_H z_{t-1}. \tag{9}$$

The return equation for the test portfolio is

$$R_t^e = \alpha + \beta_{M,t-1} MKT_t + \beta_{H,t-1} HML_t + \varepsilon_t. \tag{10}$$

Substitute (9) into (10) and expand step by step:

$$R_t^e = \alpha + (\beta_M + \delta_M z_{t-1}) MKT_t + (\beta_H + \delta_H z_{t-1}) HML_t + \varepsilon_t \tag{11}$$

$$= \alpha + \beta_M MKT_t + \beta_H HML_t + \delta_M (z_{t-1} MKT_t) + \delta_H (z_{t-1} HML_t) + \varepsilon_t. \tag{12}$$

Equation (12) is the interaction regression form. Regress R_t^e on MKT_t, HML_t, and the interaction regressors $z_{t-1} MKT_t$ and $z_{t-1} HML_t$. Assume

$$\alpha = 0, \quad \beta_M = 1.00, \quad \beta_H = 0.40, \quad \delta_M = 0.50, \quad \delta_H = -0.20.$$

Suppose there are three months of data (returns in decimals):

t	z_{t-1}	MKT_t	HML_t
1	0.0	0.020	0.010
2	1.0	−0.010	0.020
3	2.0	0.030	−0.010

Using (9), I compute time-varying betas:

$$\beta_{M,0} = 1.00 + 0.50(0) = 1.00, \qquad \beta_{H,0} = 0.40 - 0.20(0) = 0.40,$$

$$\beta_{M,1} = 1.00 + 0.50(1) = 1.50, \qquad \beta_{H,1} = 0.40 - 0.20(1) = 0.20,$$

$$\beta_{M,2} = 1.00 + 0.50(2) = 2.00, \qquad \beta_{H,2} = 0.40 - 0.20(2) = 0.00.$$

As z rises, market exposure increases while value exposure decreases. To compute model-predicted excess returns, set $\varepsilon_t = 0$. Using (10), with $\alpha = 0$,

$$\widehat{R}_t^e = \beta_{M,t-1} MKT_t + \beta_{H,t-1} HML_t.$$

Compute month by month:

$$\begin{aligned}
\widehat{R}_1^e &= 1.00(0.020) + 0.40(0.010) = 0.020 + 0.004 = 0.024 \\
\widehat{R}_2^e &= 1.50(-0.010) + 0.20(0.020) = -0.015 + 0.004 = -0.011 \\
\widehat{R}_3^e &= 2.00(0.030) + 0.00(-0.010) = 0.060 + 0 = 0.060.
\end{aligned}$$

I verify the interaction-regression calculation for $t = 2$. From (12), the fitted value at $t = 2$ is

$$\widehat{R}_2^e = \beta_M MKT_2 + \beta_H HML_2 + \delta_M (z_1 MKT_2) + \delta_H (z_1 HML_2).$$

Plug in $z_1 = 1$, $MKT_2 = -0.010$, $HML_2 = 0.020$ to get

$$\begin{aligned}
\beta_M MKT_2 &= 1.00(-0.010) = -0.010 \\
\beta_H HML_2 &= 0.40(0.020) = 0.008 \\
\delta_M (z_1 MKT_2) &= 0.50(1 \cdot -0.010) = -0.005 \\
\delta_H (z_1 HML_2) &= -0.20(1 \cdot 0.020) = -0.004
\end{aligned}$$

Summing terms gives

$$\widehat{R}_2^e = -0.010 + 0.008 - 0.005 - 0.004 = -0.011,$$

which matches the time-varying beta calculation above.

To estimate (7) by OLS, create a design matrix with columns for each factor and each interaction. For example, with two factors and one instrument, the regressors are

$$\big[\, MKT_t,\ HML_t,\ z_{t-1} MKT_t,\ z_{t-1} HML_t \,\big].$$

With FF5 and p instruments, include 5 factor columns plus $5p$ interaction columns. After estimating the coefficients, compute fitted betas over time using (8).

8.2 Latent Time-Varying Betas: State-Space Model and Kalman Filter

Instrumented betas make $\beta_{i,t}$ a deterministic function of observed instruments. A different approach treats $\beta_{i,t}$ as an unobserved state that evolves stochastically over time. This is particularly useful when exposures drift for reasons not well captured by a small set of instruments.

To streamline notation, in this subsection I suppress the asset index i and write $y_t \equiv R^e_{i,t}$ and $\beta_t \equiv \beta_{i,t}$.

8.2.1 Measurement and State Equations

Let $f_t \in \mathbb{R}^K$ be the observed factor vector at time t (for FF5, $K = 5$). The measurement equation links returns to betas:

$$y_t = \alpha + f_t^\top \beta_t + \varepsilon_t. \tag{13}$$

I assume the measurement noise is Gaussian and serially uncorrelated:

$$\varepsilon_t \mid \mathcal{I}_{t-1} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad \mathbb{E}[\varepsilon_t \varepsilon_s] = 0 \text{ for } t \neq s. \tag{14}$$

The state equation governs the evolution of betas. A standard choice is a random walk:

$$\beta_t = \beta_{t-1} + u_t. \tag{15}$$

Further assume that the state innovation u_t is Gaussian, serially uncorrelated, and independent of ε_t:

$$u_t \mid \mathcal{I}_{t-1} \sim \mathcal{N}(0, Q), \qquad \mathbb{E}[u_t u_s^\top] = 0 \text{ for } t \neq s, \qquad \mathbb{E}[u_t \varepsilon_s] = 0 \ \forall s, t. \tag{16}$$

Here $Q \in \mathbb{R}^{K \times K}$ is positive semidefinite and controls how rapidly betas can vary. A mean-reverting alternative to the random-walk formulation is the AR(1)

$$\beta_t = \bar{\beta} + \Phi(\beta_{t-1} - \bar{\beta}) + u_t, \qquad u_t \sim \mathcal{N}(0, Q),$$

where Φ is a $K \times K$ matrix with eigenvalues less than 1 in modulus. The Kalman filter recursions below are easily adapted to this case by modifying the prediction step.

8.2.2 Kalman Filter

The Kalman filter produces $\hat{\beta}_{t|t}$, the estimate of β_t using data up to time t, as well as its covariance matrix $P_{t|t}$. It proceeds recursively through a predict step and an update step, using

$$\hat{\beta}_{t-1|t-1} \quad \text{and} \quad P_{t-1|t-1}.$$

known at time $t - 1$.

1. Predict State: From (15), the conditional expectation of β_t given $\mathcal{I}_{t-1}$ is

$$\hat{\beta}_{t|t-1} \equiv \mathbb{E}[\beta_t \mid \mathcal{I}_{t-1}] = \mathbb{E}[\beta_{t-1} + u_t \mid \mathcal{I}_{t-1}] = \hat{\beta}_{t-1|t-1} + \mathbb{E}[u_t \mid \mathcal{I}_{t-1}] = \hat{\beta}_{t-1|t-1},$$

since $\mathbb{E}[u_t \mid \mathcal{I}_{t-1}] = 0$. Next compute the prior covariance. Using $\beta_t = \beta_{t-1} + u_t$ and independence of β_{t-1} and u_t conditional on $\mathcal{I}_{t-1}$,

$$\begin{aligned} P_{t|t-1} &\equiv \mathrm{Var}(\beta_t \mid \mathcal{I}_{t-1}) = \mathrm{Var}(\beta_{t-1} + u_t \mid \mathcal{I}_{t-1}) \\ &= \mathrm{Var}(\beta_{t-1} \mid \mathcal{I}_{t-1}) + \mathrm{Var}(u_t \mid \mathcal{I}_{t-1}) \\ &= P_{t-1|t-1} + Q. \end{aligned} \tag{17}$$

2. Predict Return and Compute Innovation: Given (13), the one-step-ahead predicted return is

$$\hat{y}_{t|t-1} \equiv \mathbb{E}[y_t \mid \mathcal{I}_{t-1}] = \mathbb{E}[\alpha + f_t^\top \beta_t + \varepsilon_t \mid \mathcal{I}_{t-1}] = \alpha + f_t^\top \hat{\beta}_{t|t-1},$$

because $\mathbb{E}[\varepsilon_t \mid \mathcal{I}_{t-1}] = 0$. The innovation (forecast error) is

$$\nu_t \equiv y_t - \hat{y}_{t|t-1}.$$

3. Innovation Variance: Using $y_t = \alpha + f_t^\top \beta_t + \varepsilon_t$, write

$$\nu_t = f_t^\top (\beta_t - \hat{\beta}_{t|t-1}) + \varepsilon_t.$$

Therefore,

$$\begin{aligned} S_t &\equiv \mathrm{Var}(\nu_t \mid \mathcal{I}_{t-1}) \\ &= \mathrm{Var}\Big(f_t^\top (\beta_t - \hat{\beta}_{t|t-1}) + \varepsilon_t \mid \mathcal{I}_{t-1}\Big) \\ &= \mathrm{Var}\Big(f_t^\top (\beta_t - \hat{\beta}_{t|t-1}) \mid \mathcal{I}_{t-1}\Big) + \mathrm{Var}(\varepsilon_t \mid \mathcal{I}_{t-1}) \\ &= f_t^\top P_{t|t-1} f_t + \sigma_\varepsilon^2. \end{aligned} \tag{18}$$

4. Kalman Gain: The Kalman gain is defined as

$$K_t \equiv P_{t|t-1} f_t \, S_t^{-1}. \tag{19}$$

It determines how much the state estimate responds to the innovation ν_t.

5. Update State: The updated estimate is

$$\hat{\beta}_{t|t} = \hat{\beta}_{t|t-1} + K_t \nu_t. \tag{20}$$

The updated covariance is

$$P_{t|t} = P_{t|t-1} - K_t S_t K_t^\top. \tag{21}$$

Note the dimensions. For FF5, $K = 5$, so β_t is 5×1, $P_{t|t-1}$ is 5×5, f_t is 5×1, S_t is a scalar, and K_t is 5×1.

8.2.3 Scalar Case

For intuition, consider the case of $K = 1$ and write the factor as a scalar f_t and the beta as a scalar β_t. Then

$$y_t = \alpha + \beta_t f_t + \varepsilon_t, \qquad \beta_t = \beta_{t-1} + u_t,$$

with $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ and $u_t \sim \mathcal{N}(0, q)$, the recursions are now

$$\begin{aligned} \hat{\beta}_{t|t-1} &= \hat{\beta}_{t-1|t-1}, \\ P_{t|t-1} &= P_{t-1|t-1} + q, \\ \nu_t &= y_t - \alpha - f_t \hat{\beta}_{t|t-1}, \\ S_t &= f_t^2 P_{t|t-1} + \sigma_\varepsilon^2, \\ K_t &= \frac{P_{t|t-1} f_t}{f_t^2 P_{t|t-1} + \sigma_\varepsilon^2}, \\ \hat{\beta}_{t|t} &= \hat{\beta}_{t|t-1} + K_t \nu_t, \\ P_{t|t} &= P_{t|t-1} - K_t^2 S_t. \end{aligned}$$

8.2.4 Implementation

To implement the Kalman filter for a single test portfolio, proceed as follows. First, choose the model hyperparameters $(\alpha, \sigma_\varepsilon^2, Q)$ (or estimate them by maximum likelihood). Next, initialize $\hat{\beta}_{0|0}$ and $P_{0|0}$. Then, for each $t = 1, \ldots, T$, compute the prediction $(\hat{\beta}_{t|t-1}, P_{t|t-1})$, compute the innovation ν_t and its variance S_t, compute the gain K_t, and then update to $(\hat{\beta}_{t|t}, P_{t|t})$. The output is a time series of estimated betas $\{\hat{\beta}_{t|t}\}$.

In practice, Q controls the smoothness of the resulting beta series. A small Q forces betas to change slowly over time, while a larger Q allows betas to move more quickly in response to the data. Choosing Q is therefore critical to the empirical behavior of the model, and it is common to estimate it by maximizing the Gaussian likelihood implied by the innovations ν_t.

9 Conclusion

I derive the Fama–French five-factor model starting from the consumer's intertemporal optimization problem and the associated Euler equation. By introducing the stochastic discount factor and approximating it with a linear function of traded factor returns, I obtain the familiar linear factor pricing model. Finally, I extended the framework to allow for time-varying factor loadings using both observable instruments and latent state-space methods, and provide illustrative examples.

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